Heredity (Set B) — MCQ Worksheet

Science • Chapter 8 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. A gene is a:

(A) Unit of inheritance carrying information for one trait
(B) Part of cell membrane
(C) Type of protein
(D) Lipid molecule

Q2. Different forms of the same gene are called:

(A) Alleles
(B) Chromosomes
(C) Loci
(D) Mutations

Q3. Mendel chose pea plant because it:

(A) Has clearly distinguishable traits and short life cycle
(B) Has many seasons
(C) Is large
(D) Is rare

Q4. In Mendel's cross of pure tall (TT) × pure short (tt), F1 is:

(A) All tall (Tt)
(B) All short
(C) Half tall, half short
(D) Mixed traits

Q5. Allowing F1 plants to self-pollinate (Tt × Tt) gives F2 phenotype ratio:

(A) 3 tall : 1 short
(B) 1 : 1
(C) 9:3:3:1
(D) All tall

Q6. Genotypic ratio in F2 of monohybrid cross (Tt × Tt) is:

(A) 1 TT : 2 Tt : 1 tt
(B) 3 : 1
(C) 9:3:3:1
(D) 1:1

Q7. Phenotypic ratio in F2 of dihybrid cross is:

(A) 9:3:3:1
(B) 3:1
(C) 1:2:1
(D) 1:1:1:1

Q8. Trait that does not appear in F1 generation is called:

(A) Recessive
(B) Dominant
(C) Co-dominant
(D) Incomplete dominant

Q9. Mendel's law of independent assortment is demonstrated in:

(A) Dihybrid cross
(B) Monohybrid cross
(C) Test cross
(D) Reciprocal cross

Q10. Total number of chromosomes in human gametes:

(A) 23
(B) 46
(C) 44
(D) 22

Q11. Sex chromosomes of human males:

(A) XY
(B) XX
(C) X
(D) YY

Q12. Sex chromosomes of human females:

(A) XX
(B) XY
(C) Y
(D) XO

Q13. Probability of having a male child:

(A) 1/2
(B) 1/3
(C) 1/4
(D) 2/3

Q14. Acquired traits are not inherited because:

(A) They do not change DNA of germ cells
(B) They are too weak
(C) Genes do not exist
(D) They are recessive

Q15. Variations help in:

(A) Survival when environment changes
(B) Killing other species
(C) Identical population
(D) Reducing population

Q16. If both parents have blood groups B (with genotype IBi each), what blood groups can their children have?

(A) B and O
(B) B only
(C) A and B
(D) A, B, AB and O

Q17. Probability that two parents Bb × Bb (B brown dominant) produce a brown-eyed child is:

(A) 3/4
(B) 1/4
(C) 1/2
(D) 1

Q18. Tall pea plant (Tt) crossed with short (tt). F1 phenotype is:

(A) 1 tall : 1 short
(B) 3 tall : 1 short
(C) All tall
(D) All short

Q19. If both parents are heterozygous (Aa), probability of homozygous recessive offspring (aa) is:

(A) 1/4
(B) 1/2
(C) 3/4
(D) 1

Q20. Colour blindness in humans is sex-linked recessive on X chromosome. A father (colour-blind) marries a normal woman who is not a carrier. Daughters will be:

(A) All carriers (heterozygous, normal vision)
(B) All colour-blind
(C) Half colour-blind
(D) All normal homozygous

Q21. Sons of a colour-blind father and a non-carrier mother:

(A) Are normal (no X from father)
(B) Are all colour-blind
(C) Half are colour-blind
(D) All carriers

Q22. A pure-breeding tall, red-flowered pea is crossed with pure-breeding short, white-flowered. F1 dihybrid selfed gives F2 ratio (tall red : tall white : short red : short white):

(A) 9:3:3:1
(B) 3:1
(C) 1:1:1:1
(D) 1:2:1

Q23. Test cross is used to determine:

(A) Genotype of an organism showing dominant phenotype (homozygous or heterozygous)
(B) Phenotype
(C) Cell number
(D) Cell size

Q24. When two pure-breeding red-flowered four o'clock plants are crossed with pure-breeding white, F1 is all pink (incomplete dominance). F2 phenotypic ratio is:

(A) 1 red : 2 pink : 1 white
(B) 3 red : 1 white
(C) 9:3:3:1
(D) All pink

Q25. Sex of human child is determined by:

(A) Father (X or Y from sperm)
(B) Mother
(C) Environment
(D) Diet

— End of Worksheet —